

Optipeople DemoCNC D-2800M — Technical Specifications

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1. Machine description

The Optipeople DemoCNC D-2800M is a 6-spindle CNC machining center designed for the wood and panel industry, with a focus on cabinet doors, drawer fronts, MDF panels, and solid wood components up to 2,800 mm length. It combines vertical milling with automatic tool change, manual vertical milling, horizontal hinge/lock-hole drilling and side drilling, and two bevel spindles for 45° Lamello and biscuit operations.

The machine has two alternating work zones, an auto movable cross beam, 6 top pressers, and an automatic back positioning bar. Control is via an industrial PC running the OptiPanel HMI on Windows 11 IoT LTSC, with support for online 3D editing, MPR/DXF/CSV/BAN/XML/EXCEL import, and barcode-scanned job recall.

2. Working envelope

Parameter	Value
Max workpiece length (X)	2,800 mm (extended frame to 3,000 mm with reduced clamping)
Workpiece width (Y)	80–800 mm (width ≥ 80 mm required for vacuum hold)
Workpiece thickness (Z)	12–60 mm
Min workpiece footprint	250 × 200 mm
Drilling depth, max	60 mm
Working table size	2,660 × 420 mm
Working table height	980 mm from floor
Movable beam stroke	0–160 mm
Number of work zones	2 (alternating)

3. Axis system

Axis	Drive	Rapid traverse	Repeatability	Positioning accuracy
X	Helical rack & pinion, iDIN6 class	90 m/min	±0.05 mm	±0.10 mm / 1,000 mm
Y	Pre-tensioned ball screw, C5 class	90 m/min	±0.03 mm	±0.05 mm / 1,000 mm
Z	Ball screw, C5 class	20 m/min	±0.02 mm	±0.05 mm / 1,000 mm

Drives are AC servos with absolute encoders (24-bit). The X axis is driven by a 3.0 kW servo with planetary gearbox (ratio 10:1). Y and Z axes use 1.5 kW servos.

4. Processing unit

#	Spindle	Power	Speed	Tool change	Notes
S1	Vertical, automatic tool change	7.5 kW	18,000 rpm	ATC, ISO30, ER32	Handle slots, hinge slots, decorative grooves
S2	Vertical, manual tool change	7.5 kW	18,000 rpm	Manual	Backup vertical operations, long-run tools
S3	Horizontal milling	6.0 kW	18,000 rpm	Manual collet	Top and bottom hinge cups, lock pockets
S4	Horizontal drilling	3.5 kW	18,000 rpm	Manual collet	Side drilling
S5	Bevel (45°)	2.2 kW	12,000 rpm	Manual collet	Lamello bevel groove
S6	Bevel (45°)	2.2 kW	12,000 rpm	Manual collet	Lamello holes / biscuit slots

All spindles run on 380 V 3-phase via dedicated Delta MS300 series VFDs. Spindle bearings are angular contact, lifetime-greased, with built-in PT100 temperature monitoring.

4.1 Tool magazine

Parameter	Value
Type	Inline pickup, traveling with vertical head
Slots	8
Holder	ISO30
Collet	ER32
Max tool diameter	100 mm
Max tool length	140 mm
Max tool weight	1.5 kg
Tool-to-tool change time	6.5 s
Chip-to-chip time, typical	10–12 s

4.2 Tool measurement

- 1× vertical laser tool setter (Renishaw-class, fabricated PN: OP-TS-V1)
- 2× horizontal touch probes for S3/S4 setting (PN: OP-TS-H1, OP-TS-H2)
- Auto length & diameter calibration accuracy: ± 0.01 mm

5. Workholding

Item	Detail
Table material	High-density phenolic with replaceable HPL surface
Vacuum zones	12 (6 per work zone), independently valved
Side reference bar	Pneumatic pop-up, repeatability ± 0.05 mm
Top pressers	6× pneumatic, force 2,800 N each at 6 bar
Movable beam force	3-stage cylinder, 0–160 mm stroke

Vacuum supply must be 600 m³/h minimum, supplied by external vacuum pump (not included).
Recommended: dry-running rotary vane, 5.5 kW.

6. Pneumatics

Parameter	Value
Supply pressure	6.0 bar (min 5.5, max 8.0)
Air consumption, peak	380 NL/min
Air quality	ISO 8573-1 class 4-4-3 minimum
Filtration	5 µm, 0.3 µm coalescing, on machine
Input fitting	G 1/2" female

7. Electrical

Parameter	Value
Supply voltage	380 V AC, 3-phase + N + PE
Frequency	50 Hz (60 Hz available on request)
Total connected load	33.67 kW
Recommended branch breaker	80 A, type D, IΔn 30 mA Type B RCD
Power factor (typ)	0.88
Standby load	~0.8 kW

Internal logic uses 24 V DC. The main control cabinet (PN: OP-CAB-2800) is IP54, mounted on the operator-side end of the frame, with 230 V single phase service outlet for the IPC and a separate 24 V DC ELV bus.

8. Dust extraction

Port	Diameter	Required flow
Main spindle hood (×2)	Ø 200 mm	2,200 m ³ /h each
Drilling head hood	Ø 150 mm	1,400 m ³ /h
Total	—	~5,800 m ³ /h at 2,500 Pa

9. Control system

- **IPC:** OptiPanel IPC-15 (Intel Core i5-12500T, 16 GB RAM, 512 GB SSD)
- **Display:** 15.6" projected-capacitive touchscreen, 1920×1080
- **OS:** Windows 11 IoT LTSC 24H2
- **Motion controller:** OptiMotion MC-6X (EtherCAT, 250 µs cycle, 8 axes)
- **HMI:** OptiPanel software v7.4.x
- **PLC:** Beckhoff TwinCAT 3.1, fail-safe Safety over EtherCAT (FSOE)
- **Network:** 2× 1 Gb Ethernet (one isolated for shop floor)
- **Input:** USB-A ×4, USB-C ×1, barcode scanner Y-cable
- **Remote support:** TeamViewer Tensor (whitelisted to optipeople.dk)

9.1 Supported file formats

- Native: .opt (OptiPanel job)
- Import: MPR, MPR2 (Homag), DXF, CSV, BAN, XML, XLSX
- Export: NC (G-code, ISO 6983 dialect), PDF report, CSV log

10. Safety

Item	Detail
Standards	EN ISO 12100, EN ISO 13849-1 PL d, EN 60204-1
Guarding	Full perimeter fence with 3 interlocked access doors
Door switches	Pilz PSEN-cs, coded magnetic (PN: OP-SF-D1..D3)
E-stop	4 mushroom-button stations, dual-channel
Light curtains	Optional, finger-class type 4, 14 mm resolution
Brake test	Auto on power-up (Z gravity brake)
Stop category	Cat 1 on E-stop, Cat 2 on door open during cycle

11. Physical

Parameter	Value
Machine length (L)	5,820 mm
Machine depth (W)	2,520 mm
Machine height (H)	2,000 mm
Loading height	980 mm
Total weight	2,200 kg
Floor anchor points	8× M16 chemical anchor
Floor flatness	< 3 mm over machine length
Floor loading	Min 1,500 kg/m ²

12. Environmental

Parameter	Value
Ambient temperature, operating	+5 to +40 °C
Storage temperature	-20 to +60 °C
Humidity, non-condensing	30 to 85 % RH
Altitude	< 1,000 m without derating
Vibration class (ISO 10816)	< 2.8 mm/s RMS at the foot
Noise level, idle	68 dB(A) @ 1 m
Noise level, cutting	86 dB(A) @ 1 m (hearing protection required)

13. Options

- OP-OPT-DCB — Dust removal belt under the work zones
- OP-OPT-HHS — Hinge hole spindle group (adds 5× 3.0 kW horizontal drills)
- OP-OPT-LC2 — Type-4 light curtains in lieu of physical doors
- OP-OPT-VAC — Integrated 5.5 kW vacuum pump skid
- OP-OPT-BC2 — Industrial 2D barcode scanner with stand
- OP-OPT-IO — Modbus TCP I/O coupler for MES/ERP integration

14. Warranty

3-year limited mechanical, 1-year electrical, 1-year wear parts. Spindles covered 12 months / 4,000 spindle-hours, whichever first. See contract T&Cs for exclusions.